TEAM CASE STUDY: LAUREN MITCHELL

The whole class is ONE nursing team for one patient across 24 hours. We move through the time together, 4 hours at a time.

4 West, Room 4W-22 | Mon 0700 to Tue 0700

Write your name in the header at the top of this page (it repeats on every page). Everyone fills in their own packet.

QUICK PATHOPHYSIOLOGY: ADDISON DISEASE AND ADRENAL CRISIS

The adrenal glands: The outer layer (cortex) makes **cortisol** and **aldosterone**. In **Addison disease** (primary adrenal insufficiency) the immune system destroys the cortex, so the person replaces both hormones with pills every day.

Cortisol: The stress hormone. It keeps blood pressure up (vessels respond to epinephrine), keeps blood glucose up between meals and helps the body get through illness. **Too little:** low BP, low glucose, weakness, nausea, vomiting, belly pain, confusion.

Aldosterone: Makes the kidneys keep sodium and water and get rid of potassium. **Too little:** sodium and water are lost in the urine (dehydration, low sodium) and potassium builds up (high potassium, peaked T waves, dysrhythmias).

ACTH: The pituitary keeps sending ACTH to glands that cannot answer. High ACTH darkens the skin (tan skin, dark skin creases and gums).

Adrenal crisis: Illness, injury, surgery or vomiting raises the need for cortisol, and the person cannot make more or keep the pills down. Result: **shock that fluid alone does not fix**, low glucose, low sodium, high potassium. It can kill within hours.

Treatment: **IV hydrocortisone right away** (never wait for tests; every dose is time-critical), IV saline with dextrose, treat the trigger, cardiac monitoring. High-dose hydrocortisone also acts like aldosterone, so fludrocortisone is held while the dose is high. Raise the sodium slowly (no more than 8 in 24 hours) to protect the brain.

Nursing focus: BP and orthostatic vitals, heart rate and rhythm, glucose, sodium and potassium, mental status, intake and output, fall safety, steroid doses on time, and sick-day teaching before going home.

HOW THE TWO HOURS WORK

- The instructor moves the clock forward **4 hours at a time**: 0700, 1100, 1500, 1900, 2300, 0300, and a final check at 0700 Tuesday.
- At each checkpoint, the instructor sets the bedside monitor and reads the situation. Fill in the **blanks in the situation box** from the chart. Each pair does its job in the chart and at the bedside and writes its findings in its own box, **with your names**. You decide what to look for and what to ask.
- When a pair is finished, the **charge nurse gets the findings from that pair**. The **Vitals & Assessment** and **Meds and Orders** pairs work together on the **actions to suggest**. The charge nurse decides the **top 3 priorities**, then develops and gives the **SBAR** to the provider. Everyone writes them down.
- Every stop, the **Vitals & Assessment pair** fills one column of the **response tracker** (page 3): is the treatment working? At the end you use it to **evaluate** the whole treatment.
- Finish each page at the **end** of its time block. **Do not turn the page** until the instructor advances the clock. Use the EHR on your own device (vitals, MAR, results, notes); your patient and the time are synced with the instructor. **Vitals row:** read the bedside monitor, take your own set, do orthostatic vitals when ordered and safe, and ask the instructor for the bedside glucose. Write in dark pen.

ROLES

CHARGE NURSE (one person; changes every stop)	Runs the section and keeps time. **Gets the findings from each pair when the pair is finished.** Decides the top 3 priorities with the team (writes their name in that box). **Develops and gives the SBAR.**
NURSE PAIRS (switch jobs every stop)	The other students work in pairs. **At every checkpoint each pair is assigned a new job** and writes its names in that box. **Meds and Orders:** MAR, orders, notes: what is running, due, held or should change, and why. **Vitals & Assessment:** the bedside monitor, orthostatic vitals, the bedside exam (mental status, skin, mouth, belly), intake and output. **Results & Labs:** new labs, ECG, trends and what they mean. Report to the charge nurse when you are finished.

CHARGE NURSE ROTATION (each student is charge nurse exactly once)

Stop	0700	1100	1500	1900	2300	0300	Final 0700
Charge nurse name							

TURN-IN

Each student turns in their own packet, fastened with a **paperclip (no staples)**. The last page is your individual reflection and peer check.

Turn in with a paperclip (no staples). Write in dark pen. Do not turn the page until the instructor advances the clock.

PATIENT CARD: LAUREN MITCHELL

Room 4W-22

Name: Lauren Mitchell, she/her, 46 years old, DOB 05/14/1980 | **MRN** SIM-40422 | 165 cm, **58.2 kg** on arrival (usual 61.0 kg)

ALLERGY: **Sulfamethoxazole-trimethoprim** (hives) | **Full Code** | Fall risk: HIGH | **Contact (enteric) precautions**: soap and water

History: **Addison disease** (autoimmune primary adrenal insufficiency) since 2021; Hashimoto hypothyroidism (2016); appendectomy at 19. Middle school science teacher.

Home meds: **Hydrocortisone** 15 mg at 0700 + 5 mg at 1400 (20 mg/day); **fludrocortisone** 0.1 mg daily; levothyroxine 88 mcg at 0600; vitamin D3 2,000 units daily.

Family: Husband **Daniel Mitchell** (48, health care proxy); son **Ethan** (15, had a "stomach bug" on Friday).

Care team: Day hospitalist Dr. R. Delacroix (0700-1900); night hospitalist Dr. K. Osei; endocrinology Dr. M. Harrington; pharmacist T. Nakagawa; dietitian E. Lindqvist.

Normal values: Sodium 136-145 | potassium 3.5-5.1 | glucose 70-110 | creatinine 0.5-1.1 (her usual 0.8) | cortisol (morning) 5-23 | ACTH 7.2-63

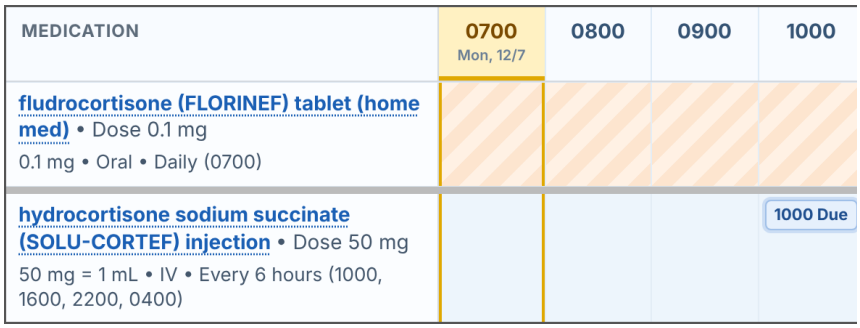
HORMONE MAP: WHY DOES SHE LOOK LIKE THIS?

fill it in at 0700

The adrenal glands make **cortisol** and **aldosterone**. In Addison disease both are missing. Write what you find in Lauren's chart and at the bedside (with numbers) for each hormone.

Hormone	What it normally does	What Lauren shows (chart numbers and bedside findings)
CORTISOL (missing)	Keeps blood vessels tight and BP up; makes glucose in the liver between meals; needed for any stress or illness; helps the kidney get rid of water.	
ALDOSTERONE (missing)	Makes the kidney keep sodium and water and get rid of potassium (and acid).	
ACTH (very high)	The pituitary pushes harder on a gland that cannot answer. ACTH comes from the same parent molecule as the skin-darkening hormone.	

FINDING IT IN THE EHR (this is what the chart shows at 0700)



MAR: the home fludrocortisone (and the home hydrocortisone, not shown) are **held**: the row is hatched. The IV hydrocortisone row shows its schedule and the next due time. Click a medicine name for the full order.

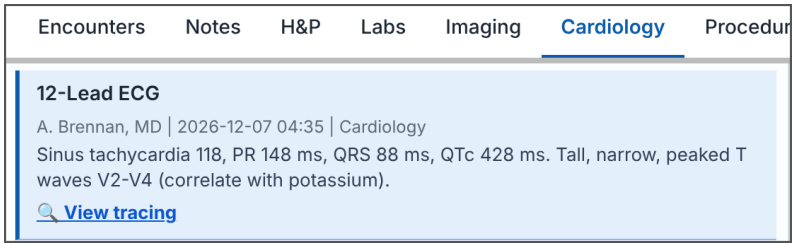


Chart Review > Cardiology: the ED 12-lead ECG. Click **View tracing** to see the strip and look at the T waves. Later ECGs appear here too.

HOW YOU EARN YOUR 20 POINTS

team grade

A complete, careful, team-minded effort earns full credit. Nothing here is a trick.

Category and points	FULL credit	GOOD (partial)	GETTING THERE
Teamwork and charge-nurse rotation **5 points**	**5.** A charge nurse is named at every stop, everyone was charge nurse once, and each pair reported its job when the charge nurse asked.	**3 to 4.** The rotation was followed most of the time. Most teammates took part.	**1 to 2.** The rotation was skipped, or one person did most of the work.
Clinical thinking at each checkpoint **6 points**	**6.** Every checkpoint has the pair boxes filled with real chart numbers and a reason (why it happens, and is the treatment working?).	**4 to 5.** Most checkpoints are filled in with numbers. One or two are thin.	**1 to 3.** Many boxes are blank or have one-word answers.
Safety, priorities and SBAR **5 points**	**5.** At 1100, 1500, 1900, 2300 and 0300: three priorities in order, a complete SBAR (all four parts) and a next step that keeps Lauren safe.	**3 to 4.** Most stops have priorities and an SBAR. A part is sometimes missing.	**1 to 2.** Priorities or SBAR are missing at many stops.
Completeness and reflection **4 points**	**4.** All boxes tried, the response tracker is filled at every stop, the final evaluation is done, and each person handed in a reflection.	**2 to 3.** Most boxes tried. The tracker, the evaluation or a reflection is missing.	**1.** Many blanks. Little reflection.

RESPONSE TRACKER: IS THE TREATMENT WORKING?

Vitals & Assessment pair fills this in

Vitals & Assessment pair: fill in this stop's column at every stop (ask the Results & Labs pair for the lab numbers). Add an arrow (up, down, same) when it helps. In the last row circle **B**etter, **S**ame or **W**orse.

Measure	Goal	Mon 0700	Mon 1100	Mon 1500	Mon 1900	Mon 2300	Tue 0300	Tue 0700
BP lying (MAP)	SBP 100 or more; MAP 65 or more							
BP and HR standing	Negative: SBP drop under 20, HR rise under 30, no dizziness							
Heart rate / T waves	Under 100; T waves normal							
Glucose (bedside)	70-180 without dextrose pushes							
Sodium (latest)	Rising, but no more than 8 in 24 h (from 126 at 0406)							
Potassium (latest)	3.5-5.1							
Mental status	Alert, oriented x4 (her baseline)							
Urine output	0.5 mL/kg/h or more (about 30 mL/h); tell if over 250 mL/h							
Nausea / eating	No nausea; eating and drinking							
OVERALL	Better, same or worse than the last stop?	B S W	B S W	B S W	B S W	B S W	B S W	B S W

0700 | Monday

Checkpoint 1 of 6

Charge nurse for this stop: _____ ☐ every pair reported to the charge nurse ☐ SBAR given

SITUATION It is **Monday 0700**, 4 West, room 4W-22. Lauren Mitchell, age _____, just came up from the ED. Her history: _____. Since Saturday she has had _____ and could not keep down her _____ pills.

EMS found her confused on the bathroom floor at 0318: BP _____, glucose _____. EMS and the ED gave _____ L of saline, dextrose, and hydrocortisone _____ at _____. Running now: _____ mL/h. Her husband _____ is at the bedside.

Time	Temp F	HR	BP lying (MAP)	Standing BP, HR	RR	SpO2	Pain	Rhythm on the monitor	Glucose
0700									

MEDS AND ORDERS Names: _____ _____

VITALS & ASSESSMENT Names: _____ _____

RESULTS & LABS Names: _____ _____

RESPONSE TRACKER (page 3): **Vitals & Assessment pair**, fill in the Mon 0700 column ☐
Hormone map (p. 2) ☐

Compared with the ED she is: ☐ better ☐ same ☐ worse

Actions to suggest (Vitals + Meds pairs)

Top 3 priorities Charge RN: _____

1.
2.
3.

SBAR to the provider

S Situation	
B Background	
A Assessment	
R Recommend	

Prediction for the next 4 hours

Charge nurse notes

Turn in with a paperclip (no staples). Write in dark pen. Do not turn the page until the instructor advances the clock.

1100 | Monday

Checkpoint 2 of 6

Charge nurse for this stop: _____ ☐ every pair reported to the charge nurse ☐ SBAR given

SITUATION It is **1100**. The 0900 bedside glucose was _____; the 0900 serum sodium is _____ and potassium _____. Her hydrocortisone _____ was due at _____. She had a large liquid stool at 0945 and the sample went to the lab.

You walk in for the 1100 glucose check. Lauren is sweating and says, "Something's wrong." Her bedside glucose is _____ (ask the instructor). Her mental status compared with 0700: _____.

Time	Temp F	HR	BP lying (MAP)	Standing BP, HR	RR	SpO2	Pain	Rhythm on the monitor	Glucose
1100									

MEDS AND ORDERS Names: _____ _____

VITALS & ASSESSMENT Names: _____ _____

RESULTS & LABS Names: _____ _____

RESPONSE TRACKER (page 3): **Vitals & Assessment pair**, fill in the Mon 1100 column ☐ | Since the last stop she is: ☐ better ☐ same ☐ worse

What changed since the last stop?

Actions to suggest (Vitals + Meds pairs)

Top 3 priorities Charge RN: _____

1.
2.
3.

SBAR to the provider

S Situation	
B Background	
A Assessment	
R Recommend	

Prediction for the next 4 hours

Charge nurse notes

Turn in with a paperclip (no staples). Write in dark pen. Do not turn the page until the instructor advances the clock.

1500 | Monday

Checkpoint 3 of 6

Charge nurse for this stop: _____ ☐ every pair reported to the charge nurse ☐ SBAR given

SITUATION It is **1500**. Lauren has kept clear liquids down since _____, and endocrinology saw her. The ED cortisol (before the steroid) is _____ and the ACTH is _____. The 1300 sodium is _____: up _____ from 126 in _____ hours.

The urine hat keeps filling with pale urine: about _____ mL since 1100. Her lying BP is over 100 and she is alert, so **orthostatic vital signs are due**. The next hydrocortisone is due at _____.

Time	Temp F	HR	BP lying (MAP)	Standing BP, HR	RR	SpO2	Pain	Rhythm on the monitor	Glucose
1500									

MEDS AND ORDERS Names: _____ _____

VITALS & ASSESSMENT Names: _____ _____

RESULTS & LABS Names: _____ _____

- Endocrine
  - ✓ TSH
  - ✓ Free T4
  - ✓ **Cortisol, random (drawn before hydrocortisone)**
  - ✓ ACTH (drawn before hydrocortisone)

RESPONSE TRACKER (page 3): **Vitals & Assessment pair**, fill in the Mon 1500 column ☐

Since the last stop she is: ☐ better ☐ same ☐ worse

What changed since the last stop?

Actions to suggest (Vitals + Meds pairs)

Top 3 priorities Charge RN: _____

1.
2.
3.

SBAR to the provider

S Situation	
B Background	
A Assessment	
R Recommend	

Prediction for the next 4 hours

Charge nurse notes

Turn in with a paperclip (no staples). Write in dark pen. Do not turn the page until the instructor advances the clock.

1900 | Monday

Checkpoint 4 of 6

Charge nurse for this stop: _____ ☐ every pair reported to the charge nurse ☐ SBAR given

SITUATION It is **1900**, shift change: **your team now becomes the night shift**. At 1530 the provider changed the IV fluid from _____ to _____. Lauren has had ginger ale and juice, a _____ diet was ordered at 1830, and she had one loose stool at 1640.

The 1700 sodium is _____ and potassium _____. The stool GI panel found _____, so she stays on _____ precautions; clean hands with _____. Orthostatic vitals were ordered again for 1900. Tonight's hydrocortisone is due at _____ and _____.

Time	Temp F	HR	BP lying (MAP)	Standing BP, HR	RR	SpO2	Pain	Rhythm on the monitor	Glucose
1900									

MEDS AND ORDERS Names: _____ _____

VITALS & ASSESSMENT Names: _____ _____

RESULTS & LABS Names: _____ _____

COLLECTED	RESULT	FLAG
2026-12-07 04:06	126 mmol/L	Low
2026-12-07 09:00	128 mmol/L	Low
2026-12-07 13:00	132 mmol/L	Low

RESPONSE TRACKER (page 3): **Vitals & Assessment pair**, fill in the Mon 1900 column ☐

Since the last stop she is: ☐ better ☐ same ☐ worse

What changed since the last stop?

Actions to suggest (Vitals + Meds pairs)

Top 3 priorities Charge RN: _____

1.

2.

3.

SBAR to the provider

S Situation	
B Background	
A Assessment	
R Recommend	

Prediction for the next 4 hours

Charge nurse notes

Turn in with a paperclip (no staples). Write in dark pen. Do not turn the page until the instructor advances the clock.

2300 | Monday night

Checkpoint 5 of 6

Charge nurse for this stop: _____ ☐ every pair reported to the charge nurse ☐ SBAR given

SITUATION It is **2300**. Lauren ate _____% of her dinner at 2000. The 2100 sodium is _____ and potassium _____; the sodium limit until 0400 is _____. The hydrocortisone was due at _____: given? _____.

Her call light is on. "My mind won't stop. Who's going to take Ethan to school? Can I have something to knock me out?" Her bedside glucose is _____ (ask the instructor).

Time	Temp F	HR	BP lying (MAP)	Standing BP, HR	RR	SpO2	Pain	Rhythm on the monitor	Glucose
2300									

MEDS AND ORDERS Names: _____ _____

VITALS & ASSESSMENT Names: _____ _____

RESULTS & LABS Names: _____ _____

RESPONSE TRACKER (page 3): **Vitals & Assessment pair**, fill in the Mon 2300 column ☐ | Since the last stop she is: ☐ better ☐ same ☐ worse

What changed since the last stop?

Actions to suggest (Vitals + Meds pairs)

Top 3 priorities Charge RN: _____

1.
2.
3.

SBAR to the provider

S Situation	
B Background	
A Assessment	
R Recommend	

Prediction for the next 4 hours

Charge nurse notes

0300 | Tuesday

Checkpoint 6 of 6

Charge nurse for this stop: _____ ☐ every pair reported to the charge nurse ☐ SBAR given

SITUATION It is **0300 Tuesday**. At _____ the bed alarm went off: Lauren had walked to the bathroom _____, had a soft loose stool, felt lightheaded and sat down on the toilet, then pulled the cord. You found her sitting there. **She did not fall.**

She is back in bed. The 0100 potassium is _____ (it was 6.0 at 0406) and sodium _____. Look at her and at both IV sites: the right forearm IV is _____. The next hydrocortisone is due at _____.

Time	Temp F	HR	BP lying (MAP)	Standing BP, HR	RR	SpO2	Pain	Rhythm on the monitor	Glucose
0300									

MEDS AND ORDERS Names: _____ _____

VITALS & ASSESSMENT Names: _____ _____

RESULTS & LABS Names: _____ _____

RESPONSE TRACKER (page 3): **Vitals & Assessment pair**, fill in the Tue 0300 column ☐

Since the last stop she is: ☐ better ☐ same ☐ worse

What changed since the last stop?

Actions to suggest (Vitals + Meds pairs)

Top 3 priorities Charge RN: _____

1.
2.
3.

SBAR to the provider

S Situation	
B Background	
A Assessment	
R Recommend	

Prediction for the next 4 hours

Charge nurse notes

0700 | Tuesday

Final check, page 1 of 2

Charge nurse for this stop: _____ ☐ every pair reported to the charge nurse ☐ SBAR given

SITUATION It is **Tuesday 0700**. Lauren slept from about 0330. The 0500 sodium is _____ (it was 126 on arrival) and potassium _____. Endocrinology has written the plan to switch to _____ pills, and _____ restarts at _____. She wants to go home today: "What do I do next time Ethan gets sick?"

Daniel is back at the bedside: "Show me the shot. I'll be the one giving it if she can't." Her weight this morning is _____ kg (arrival 58.2 kg).

Time	Temp F	HR	BP lying (MAP)	Standing BP, HR	RR	SpO2	Pain	Rhythm on the monitor	Glucose
0700									

MEDS AND ORDERS Names: _____ _____

VITALS & ASSESSMENT Names: _____ _____

RESULTS & LABS Names: _____ _____

RESPONSE TRACKER (page 3): **Vitals & Assessment pair**, fill in the Tue 0700 column ☐

Since the last stop she is: ☐ better ☐ same ☐ worse

EVALUATE: DID THE TREATMENT WORK?

Use the response tracker (page 3) and the chart. Write the number at arrival (Mon 0700) and now, then decide: goal met?

Criterion	Goal	At arrival (Mon 0700)	Now (Tue 0700)	Goal met?
Blood pressure / MAP	SBP 100 or more, MAP 65 or more			☐ yes ☐ no
Orthostatic vitals	Negative			☐ yes ☐ no
Heart rate and T waves	Under 100; T waves normal			☐ yes ☐ no
Glucose	70-180 without dextrose pushes			☐ yes ☐ no
Sodium	Up, but no more than 8 in 24 h			☐ yes ☐ no
Potassium	3.5-5.1; normal T waves			☐ yes ☐ no
Mental status	Oriented x4 (baseline)			☐ yes ☐ no
Kidneys and urine	Creatinine back to baseline (0.8); urine 0.5 mL/kg/h or more			☐ yes ☐ no
Nausea / eating	No nausea, eating and drinking			☐ yes ☐ no

Our verdict on the treatment

Turn in with a paperclip (no staples). Write in dark pen. Do not turn the page until the instructor advances the clock.

0700 | Tuesday

Going home safely. Same charge nurse.

Final check, page 2 of 2

What changed since 0300 and since arrival?	Actions to suggest (Vitals + Meds pairs)

Top 3 priorities for the day team Charge RN: _____

1.
2.
3.

Handoff to the day team (SBAR)

S Situation	
B Background	
A Assessment	
R Recommend	

MEDS AND ORDERS: the steroid plan at home Names: _____ _____

Sick-day rules we will teach	The emergency injection

Medical ID, card and who needs to know	Our teach-back question for Lauren and Daniel

Is she ready to go home? What must be true first?

Charge nurse notes

Turn in with a paperclip (no staples). Write in dark pen. Do not turn the page until the instructor advances the clock.

MY REFLECTION AND PEER CHECK

individual

Hand this page in with your own name on it.

Name: _____ Team: _____ My charge stop: _____

1. Something I learned about adrenal crisis from Lauren's case (how the body, the treatment or the evaluation works)

2. Something our team did well (and who helped)

3. One thing I would do differently as a charge nurse or as a nurse

PEER TEAMWORK CHECK

Rate up to three teammates (not yourself). Mark one box for each.

Teammate	Did their share	Listened to teammates	Spoke up with good ideas	One kind, useful comment
	☐ yes ☐ mostly ☐ not yet	☐ yes ☐ mostly ☐ not yet	☐ yes ☐ mostly ☐ not yet	
	☐ yes ☐ mostly ☐ not yet	☐ yes ☐ mostly ☐ not yet	☐ yes ☐ mostly ☐ not yet	
	☐ yes ☐ mostly ☐ not yet	☐ yes ☐ mostly ☐ not yet	☐ yes ☐ mostly ☐ not yet	

Hand in your reflection with the team packet. Thank you!